

LLM Evaluators Represent Rank, Not Magnitude: A Signed-Budget Ballot Assay - Working

John Connor Sullivan

July 11, 2026

Abstract

Numerical scores from LLM judges need not correspond to stable internal magnitudes. We test this using *Absolute Allocation*: a signed budget ballot that elicits support, opposition, neutrality, and numerical intensity alongside best-pick and weak ordinal judgments. In a 100-prompt Qwen evaluator study, candidate-span activations linearly decode winners, pairwise ordering, and allocation polarity. Help and hurt scores recover much of each other’s decodable information, consistent with approximately opposite readouts of a shared comparative dimension. In contrast, linear probes consistently fail to predict allocation magnitude after conditioning on the generated ordinal ballot. Semantic probe directions also causally influence expressed allocations beyond matched-norm random perturbations, although the effect is boundary-dependent and most usable intervention outputs require deterministic budget normalization. The central dissociation is therefore behavioral versus representational: the elicited ballot is cardinal-looking, while the linearly readable candidate representation is principally comparative and bipolar. Reference display, position effects, and cross-model elicitation failures are reported as diagnostic limits on this assay rather than separate contributions.

1 Introduction

LLMs are often sampled multiple times and aggregated by majority vote, self-consistency, ranked voting, or judge selection [8, 7, 10]. These methods can improve reliability when errors are independent, but LLM judges also share correlated artifacts: verbosity preference, position effects, format sensitivity, style preference, and plausible but wrong reasoning. A useful voting protocol for studying LLM evaluators should therefore reveal not only which answer is preferred, but also which answers are actively opposed, ignored, or supported only weakly.

We study *Absolute Allocation voting* [4]. In each election, an evaluator sees the original prompt and four candidate answers, then casts:

1. a single best-pick vote,
2. a Borda-style ordinal ballot,
3. an Absolute Allocation ballot over 100 influence cents.

Positive cents help a candidate, negative cents hurt a candidate, and the sum of absolute cents must be exactly 100:

$$A = +60, \quad B = -40, \quad C = 0, \quad D = 0.$$

The point is not only aggregation. The richer ballot is a behavioral assay: best-pick gives a winner, the ordinal ballot gives relative order, and Absolute Allocation adds signed support, opposition, neutrality, and intensity.

We distinguish four objects throughout. *Ordinal information* is candidate ordering or pairwise comparison. *Polarity* is positive, zero, or negative allocation. *Intensity* is allocation magnitude conditional on the ordinal judgment and polarity. A *bipolar readout* means that help and hurt are consistent with approximately opposite readouts of a shared decodable dimension; it does not imply a stable cardinal utility. *Cardinal information* would require reproducible magnitude differences beyond ordinal position. Absolute Allocation elicits all four behaviorally, but whether each has a corresponding decodable representation is an empirical question.

This paper has four main claims.

1. **Linear evidence favors a bipolar comparative signal over residual cardinal intensity.** Candidate activations decode polarity, winners, and pairwise ordering, while multiple linear specifications fail to add out-of-sample value for magnitude after conditioning on the generated weak ordinal ballot.
2. **Semantic probe directions are behaviorally causal.** Their activation produces allocation movement beyond matched-norm random strength, but this establishes influence on expressed comparative judgment, not direct control of a stable latent utility.
3. **Signed ballots separate measurement from representation.** The ballot behavior contains polarity and magnitude beyond weak ranking; only part of that richer behavioral object is linearly readable at the tested activation site.
4. **Context and model family delimit assay validity.** Reference display and presentation position change Qwen votes, while Gemma and Llama do not reliably produce a directly comparable signed behavioral object.

2 Related Work

Self-consistency aggregates multiple sampled answers by majority vote [8], and recent work extends aggregation to ranked voting rules such as Borda count and instant-runoff voting [7], or exploits higher-order structure beyond majority vote [1]. Our ballot differs by eliciting a bounded *signed* influence budget, which makes opposition and neutrality first-class behavioral labels rather than derived quantities. Multi-agent debate can improve factuality but can also converge on persuasive errors [3]; our behavioral runs echo this caution, since shared debate raised voter unanimity (53% vs. 7% under private critique) without raising accuracy, while inference-time private critique [2] preserved voter diversity. The LLM-as-a-judge literature documents position and verbosity biases [10, 6]; our reference-display analysis adds self-generated reference conditioning to that catalog of artifacts, and our position analysis expresses primacy bias as graded support and opposition. Finally, representation engineering asks whether high-level attributes can be read from activations [11, 9] and steered by adding activation vectors during generation [5]; we apply both reading and steering to evaluator preference itself. Prior work largely asks whether judge evaluations can be decoded or altered; our contrastive question is whether signed numerical judgments correspond to bipolar, ordinal, or cardinal internal structure.

3 Experimental Setup

We use 100 prompts spanning factual QA, reasoning, writing, safety, code, business advice, instruction following, analysis, and ethics/policy. For each prompt, Qwen/Qwen2.5-3B-Instruct generates four candidate answers in different answer styles. Candidate order is shuffled during evaluation, and style labels are hidden.

The main evaluator is `Qwen/Qwen2.5-7B-Instruct`. For behavioral validation, we compare direct best-pick aggregation, Borda-style aggregation, Absolute Allocation aggregation, and a single weak selector. The reported validation specification uses 10 voters with private critique and a verified repeated `Qwen/Qwen2.5-14B-Instruct` external judge.

We additionally run behavioral cross-model checks with `meta-llama/Llama-3.1-8B-Instruct` and `google/gemma-3-12b-it`. These runs reuse the same 100 prompts, the same frozen set of 400 candidate answers, the same ballot prompt, four evaluator repeats, and shuffled candidate presentation. They test whether the elicited ballot behavior generalizes; they do not enter the Qwen activation probes or steering directions reported below.

For representation analysis, each evaluator first generates its own answer to the prompt. We then run two variants. In the *visible-reference* variant, that self-answer is shown during voting as a private reference. In the *hidden-reference* variant, the self-answer is generated but not shown during voting; it is used only afterward as a measurement of the evaluator’s latent answer tendency. The hidden-reference design is the cleaner mechanistic condition because it avoids direct in-context anchoring.

The primary Qwen paired experiment freezes the candidate corpus, self-answers, candidate order, and randomized ballot-field order across conditions. A hidden anchor uses seed 7; the visible treatment and hidden placebo both use seed 11. The anchor yields 383 usable ballots. Because a schema bug archived display mappings only for valid anchor ballots, 17 invalid anchor contexts are omitted from both paired conditions; analyses are therefore complete-case and conditioned on anchor validity. The visible and placebo runs yield 379 and 371 usable ballots, respectively.

The primary hidden representation dataset has 100 prompts, 383 usable ballots, and four candidates per ballot, for 1,532 prompt-evaluator-candidate rows. We extract candidate-span activations from the voting context and self-answer activations from the separate self-answer context. Probes use prompt-grouped cross-validation. The primary probe preselects layer 16 by architecture depth and replays the exact chat-formatted ballot context; feature-family and target comparisons remain exploratory.

The exact signed budget remains nontrivial: 338 of 383 usable hidden ballots are natively valid. Restricting probes to these ballots leaves activation-layer results essentially unchanged (best-pick AUC 0.750 versus 0.751 overall, signed-top AUC 0.799 versus 0.800, help 0.693 versus 0.698, hurt 0.728 versus 0.725, and allocation z score $R^2 = 0.217$ versus 0.219).

Table 1 separates the experimental versions and their units. We reserve *exact-context replay* for identical saved inputs, orders, and prompt construction. A *natively budget-valid* output parses and satisfies the absolute budget without rescaling; a *normalized* allocation has parseable signed values divided by their absolute total; *usable* includes either; and *unusable* means no complete finite allocation remains after retries. Thus “exact context” does not imply native budget validity.

Experiment	Model	Prompts	Effective unit	Purpose
Behavioral validation	Qwen	100	prompt	Aggregation sanity check
Primary probing	Qwen-7B	100	383 ballots	Representation
Reference display	Qwen-7B	100	paired prompt	Context stress test
Cross-model elicitation	Llama/Gemma	100	ballot	Measurement invariance
Activation steering	Qwen-7B	25	50 ballot clusters	Intervention

Table 1: Experiment map. Probe candidate rows and intervention generations are repeated observations; cross-validation groups by prompt and steering standard errors cluster by ballot.

4 Behavioral Measurement and Diagnostic Checks

The aggregation result is mainly a sanity check that the ballots are behaviorally meaningful. Under the verified 14B judge, 10 voters with private critique give the results in Table 2. Absolute Allocation is competitive and has the best exact match among the direct voting methods, but the margin over Borda-style aggregation is not decisive. We therefore treat aggregation as validation, not the intellectual center of the paper.

Method	n	Exact match	Tie-inclusive match
Direct best pick	100	0.34	0.49
Direct Borda-style	100	0.35	0.50
Direct Absolute Allocation	100	0.37	0.47
Single weak selector	100	0.26	0.41

Table 2: Behavioral validation under verified Qwen/Qwen2.5-14B-Instruct external judge ($n = 100$ prompts; 95% binomial intervals on these rates are roughly ± 10 points).

4.1 Reference-Answer Display Effects

The Qwen reference-display comparison reuses exact self-answers, candidate order, and ballot-field order. The same-seed visible-versus-hidden-placebo contrast changes the elected winner on 45% of prompts for best-pick, 38% for Borda-style aggregation, and 24% for Absolute Allocation. The anchor-versus-placebo comparison gives a seed-instability floor of 9%, 12%, and 10%. Separately, anchor-versus-visible change rates are 46%, 39%, and 26%, yielding net rate differences of 37, 27, and 16 percentage points. These differences are descriptive, not automatically causal corrections. Paired McNemar tests compare, prompt by prompt, the indicator that anchor-versus-visible changed the winner with the corresponding anchor-versus-placebo indicator ($p < 0.002$ for each method; $p < 10^{-9}$ for best-pick). The effect is largest for best-pick and smallest for Absolute Allocation, consistent with richer ballots producing more stable winners. On paired candidate rows, visible references lower mean allocation by 0.019, change allocation polarity on 24.3%, increase hurt from 36.8% to 42.7%, and reduce neutrality from 3.3% to 0.9%.

The Llama companion experiment isolates display more tightly. The visible run reuses the exact saved self-answer and candidate order from the hidden run. All 400 self-answer texts match; 369 ballots parse in both conditions, spanning 99 prompts. Table 3 reports winner changes after aggregating only voters parsed in both conditions. Best-pick winners change on 27.3% of prompts (95% Wilson interval 19.5–36.8%), while Borda and Absolute Allocation winners change on 11.1% and 10.1%. Restricting to the 77 prompts with all four voters paired gives nearly identical rates: 28.6%, 10.4%, and 11.7%.

Reference display also changes Llama’s allocation pattern, but not in the Qwen direction. Help rises from 34.1% to 38.4%, neutrality falls from 22.6% to 19.3%, and hurt falls slightly from 43.3% to 42.3%. Thus the visible reference shifts neutrality mainly toward support rather than making Llama more oppositional. Candidate polarity changes on 35.1% of paired rows. However, only 12 ballots are natively valid in both conditions; within those 48 candidate pairs, polarity changes on 6.25% and mean absolute allocation movement is 0.060. Exact magnitude and polarity-change claims are therefore substantially less secure than the best-pick result.

Reference-guided judging should consequently be treated as a first-order and model-dependent design choice. The Qwen comparison establishes a reference- display effect across all three ag-

Condition	Best-pick	Borda-style	Absolute Allocation
Qwen, exact inputs and same seed	45.0%	38.0%	24.0%
Qwen, seed-only placebo floor	9.0%	12.0%	10.0%
Qwen, net difference from floor (pts)	37.0	27.0	16.0
Llama, exact paired inputs ($n = 99$)	27.3%	11.1%	10.1%
Llama, four paired voters ($n = 77$)	28.6%	10.4%	11.7%

Table 3: Aggregate winner changes between hidden- and visible-reference conditions ($n = 100$ prompts). Qwen’s primary contrast compares the visible run directly with a hidden placebo at the same seed and exact paired inputs; the net-difference row subtracts the seed-instability rate from the separate anchor-versus-visible change rate. McNemar tests compare the two per-prompt change indicators rather than testing the subtraction itself ($p < 0.002$). Llama lacks a same-seed placebo but reuses exact self-answers and presentation orders and retains only ballots parsed in both conditions.

gregation rules above an estimated seed-instability floor and shows that a visible reference makes Qwen more oppositional (hurt rising from 36.8% to 42.7%, neutrality falling from 3.3% to 0.9%). The Llama comparison lacks a same-seed placebo and shifts instead toward support. These outcomes are compatible with anchoring, self-preference, consistency pressure, added information, or contrast; the design does not distinguish among them.

4.2 Position Bias

Presentation order has a strong graded effect. In the hidden-reference run, the first-shown candidate receives the best-pick vote 47.5% of the time versus 12.5% for the fourth. Mean Absolute Allocation drops from 0.202 for position 1 to 0.080 for position 4. The first-shown candidate is helped 70.0% of the time and hurt 28.5%; the fourth is helped 52.2% and hurt 43.6%. Absolute Allocation is useful here because it exposes position bias as graded support and opposition, not only as a top-choice artifact.

4.3 Cross-Model Measurement Invariance

The same prompt does not reliably elicit a comparable signed behavioral object from every model (Table 4). Llama’s parsed ballots have nearly the same help/neutral/hurt proportions as Qwen, although Llama satisfies the absolute-cent budget far less often and voluntarily returns a strict ranking on every parsed ballot. Gemma has the opposite profile: it follows the output schema and budget reliably, but usually treats Absolute Allocation as a winner-take-all vote.

Evaluator	Parsed	Native	One-hot	Help	Neutral	Hurt
Qwen2.5-7B (fixed-field experiment)	400	60.5%	9.5%	33.1%	21.9%	44.9%
Llama3.1-8B	387	14.0%	7.8%	34.0%	22.5%	43.5%
Gemma3-12B	400	97.3%	81.8%	34.0%	64.7%	1.3%

Table 4: Archived cross-model hidden-reference behavior on the same 100 prompts, candidate answers, and fixed-field ballot prompt. Native is the fraction of 400 attempted ballots requiring no repair or normalization; one-hot is among parsed ballots; polarity rates are among parsed candidate rows.

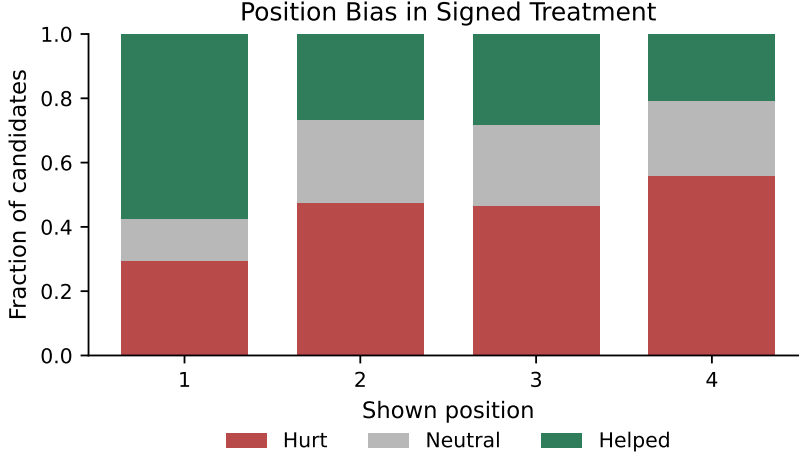


Figure 1: Position bias in the fixed-field-order hidden-reference experiment. The randomized-field primary experiment retains the best-pick primacy gradient but has a more positive allocation distribution overall.

Gemma selects candidate A on 322 of 400 ballots and gives all 100 cents to one candidate on 327 ballots. Of 389 natively valid ballots, only six candidate rows (0.39%) receive negative allocation. Even when Gemma uses tied ordinal groups, Absolute Allocation adds little information: among 201 tied candidate pairs, only 10 receive different allocations and only two differ in sign. This leaves no defensible native hurt class from which to estimate a bipolar probe or intervention direction. We therefore retain Gemma as evidence that the assay lacks measurement invariance under the current prompt, not as evidence of a different internal preference geometry, and exclude it from representation and steering analyses.

This collapse cannot yet be attributed wholly to Gemma’s internal preference geometry. The formatting example in the shared ballot prompt uses the real candidate IDs, demonstrates A as best, ranks $A > B > C > D$, and assigns A all 100 cents. Candidate order is shuffled, but candidate identity is not relabeled; Gemma’s A preference persists at every shown position. Gemma may therefore be especially susceptible to the schema example. A neutral-example or candidate-label permutation ablation is required before interpreting its winner-take-all behavior as intrinsic.

5 Representation Results

5.1 Candidate Activations Predict Ballots

Candidate activations predict the evaluator’s own voting behavior in the hidden-reference run (Table 5). Candidate activations alone carry most of the signal; self-answer-relative features add small increments, and simple self-answer geometry alone is weak. Thus the self-answer design is useful as an ablation, but the self-answer is not the main source of predictive power in these linear probes.

Several secondary checks support the same conclusion. Pairwise candidate-difference activations predict which candidate receives more Absolute Allocation with AUC 0.787. Position is predictive by itself (position-only best-pick AUC 0.683), but decodability survives matching on shown position: pooled within-position AUC is 0.659 for help, 0.719 for hurt, 0.666 for best-pick, and 0.771 for

Target	Candidate activation	+ self-answer delta	Self geometry only
Voter best-pick vote	AUC 0.772	AUC 0.774	AUC 0.530
Voter allocation top choice	AUC 0.821	AUC 0.829	AUC 0.650
Voter allocation, z-scored	$R^2 = 0.229$	$R^2 = 0.241$	$R^2 = 0.036$

Table 5: Primary randomized-context hidden-reference feature-family comparison at the preselected layer 16. The self-answer is generated before voting but not shown.

allocation top choice. Randomized ballot field order alone is at chance, and self-answer geometry gives only $R^2 = 0.036$ for allocation z score.

5.2 Absolute Allocation Refines Weak Ordinal Groups

In the fixed-field experiment, the ordinal ballot is Borda-style, but it is not canonical strict Borda. The prompt allowed ranked groups, so the observed ordinal object is a weak ranking with ties. This matters: only 17.5% of hidden-reference ballots are strict 1-1-1-1 rankings. The most common weak-ranking patterns are [1,3] at 36.75% and [1,2,1] at 36.75%.

Absolute Allocation adds signed structure inside these coarse ordinal groups. Within tied-rank candidate pairs, 47.0% receive different allocations and 35.9% receive allocations of different sign. Table 6 shows dense-rank polarity. Dense rank 2 has high hurt frequency, but because rank 2 often denotes a tied non-top group rather than strict second place, this should not be read as evidence for systematic “competitive opposition” to strict second-place candidates. In the strict-ranking subset, hurt rates increase monotonically with worse rank: 32.86% at rank 2, 50.00% at rank 3, and 61.43% at rank 4. The safer interpretation is that Absolute Allocation refines weak/coarse ordinal judgments.

Dense rank	n	Help rate	Neutral rate	Hurt rate	Mean allocation
1	400	0.980	0.020	0.000	0.584
2	841	0.137	0.244	0.620	-0.061
3	289	0.066	0.398	0.536	-0.088
4	70	0.057	0.329	0.614	-0.086

Table 6: Dense rank from the Borda-style weak ordinal ballot versus Absolute Allocation polarity in the fixed-field hidden-reference experiment. Dense ranks are not equivalent to positions in a forced strict Borda ranking.

Rank-matched probes show that some of this signed structure is visible in activations. Within dense weak-ordinal rank 2, candidate activations predict help (AUC 0.607), hurt (AUC 0.635), and neutrality (AUC 0.584), outperforming position-only baselines (0.516, 0.523, and 0.550). The effects are modest, but they show that the signed labels are not purely formatting noise.

5.3 Bipolar Comparative Readout Without Residual Linear Intensity

The key negative result is residual intensity. Borda-style weak-ordinal targets are already predictable: in the hidden run, Borda-style z score reaches $R^2 = 0.148$, compared with Absolute Allocation z score $R^2 = 0.242$. But after regressing Absolute Allocation targets on the voter’s own ordinal-ballot features, candidate activations do not predict the residuals. Across five prompt-grouped folds, residual mean $R^2 \pm \text{SD}$ is -0.036 ± 0.039 for raw allocation, -0.002 ± 0.017 for

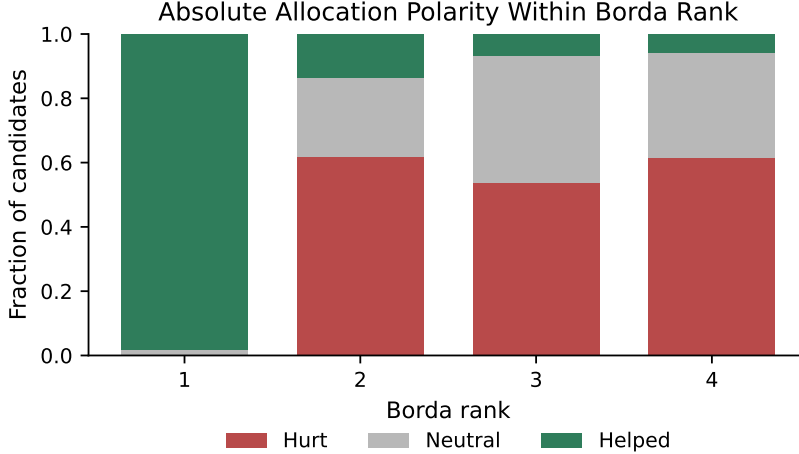


Figure 2: Absolute Allocation decomposes each dense weak-ordinal rank into helped, neutral, and hurt candidates. Dense rank 2 includes many tied non-top groups, so its high hurt rate reflects signed structure inside coarse ordinal groups, not opposition to strict second-place competitors.

centered allocation, 0.003 ± 0.017 for allocation z score, -0.040 ± 0.027 for positive mass, and -0.006 ± 0.034 for negative mass. Adding shown-position covariates does not improve the result.

This null is not trivial: the weak ordinal ballot leaves real behavioral variance, because most ballots contain ties and tied candidates often receive different signed allocations. The residual result says that this extra cardinal-looking variance yields no out-of-sample predictive value for the linear probes, layer, token pooling, and sample tested here. This is a consistent predictive failure across several intensity targets, not an equivalence test. Nonlinear, distributed, later-layer, or relational encodings remain possible, as does attenuation from noisy allocation labels or conditioning on the generated ranking.

Support and opposition, by contrast, behave as opposite ends of one decodable axis. Raw help/hurt direction cosines (about -0.91) are attenuated by low split-half coefficient reliability (0.04 at the probe’s default regularization, rising to 0.08 – 0.11 at $C = 0.01$). At $C = 0.01$ the attenuation-corrected cosine is nonetheless a stable -1.00 (50 of 50 valid split-half corrections in both standardized and raw coordinates), destabilizing only at higher regularization, so the cosine is only reliability-limited corroboration. Score sufficiency is the primary, regularization-free test. If support and opposition share a dimension, the help probe’s scalar score should carry the decodable hurt information. It does (Table 7). The help score recovers much of the decodable hurt information, and the hurt score predicts help. The AUC differences are not formally tested, so this is evidence consistent with a shared dimension, not proof of one-dimensional sufficiency. Neutrality is weak and unstable to decode in this condition, where its frequency is protocol-dependent.

6 Fresh-Baseline Causal Steering

The probing results above are observational. We therefore train help and hurt directions on the training prompts and inject them into the layer-16 residual stream only over a target candidate’s answer-span tokens during ballot generation [5]. The intervention experiment preselects the layer, selects targets from a fresh paired baseline, uses the exact randomized-field ballot prompt, and

Test (hidden run, layer 16)	AUC
Hurt from full activation space	0.659 ± 0.035
Hurt from help-probe score alone (1-D)	0.629 ± 0.040
Help from full activation space	0.620 ± 0.046
Help from hurt-probe score alone (1-D)	0.646 ± 0.046
Neutral from full activation space	0.608 ± 0.074
Neutral from help and hurt scores (2-D)	0.516 ± 0.178

Table 7: Score-sufficiency and neutrality tests (prompt-grouped 5-fold CV; mean $\pm$ std). Opposite-polarity scores recover substantial information, consistent with shared-dimensional readouts; no equivalence test is implied.

adds two additional unsteered generations to measure baseline noise. Of 80 attempted baselines, 50 yield usable allocations and enter the intervention grid. Every top target is fresh rank 1; 44 of 50 bottom targets are fresh rank 4 and six are rank 3. Across the two replicate baselines, mean absolute candidate-level allocation movement is 0.041, the best-pick flips in 7.4% of ballots, and the allocation-top choice flips in 6.3%.

The exact-budget requirement is a major qualification. Of 2,100 steered generations, 213 are natively valid, 1,660 have a parseable signed allocation that is deterministically normalized by its absolute total, and 227 remain unusable. The allocation regressions therefore use 1,873 normalized or natively valid paired outcomes across 50 ballot clusters. The primary baseline has the same issue: the 50 usable intervention baselines comprise three natively valid and 47 normalized ballots, while 30 other attempted baselines are unusable. The estimates are consequently causal effects on the recoverable-ballot subset under a normalization rule, not natively budget-valid effects.

Table 8 shows the 20% intervention. The most coherent effects occur away from a boundary: help and negated hurt promote bottom targets, while hurt and negated help demote top targets. At 20%, the two upward semantic directions move bottom targets by +0.043 and +0.052, compared with +0.018 for the three-direction random mean. The two downward semantic directions move top targets by -0.112 and -0.105, compared with -0.041 for the random mean. Effects in the ceiling- or floor-limited direction are weak and sometimes share the random drift, so the rerun does not support a uniformly symmetric axis at every target position.

Intervention at 20%	Δ Allocation	Δ Best-pick	Δ Borda
help $\rightarrow$ bottom	+0.043	+0.040	+0.291
-hurt $\rightarrow$ bottom	+0.052	+0.060	+0.330
random $\rightarrow$ bottom (3-direction mean)	+0.018	+0.013	+0.107
-help $\rightarrow$ bottom	+0.014	0.000	-0.047
hurt $\rightarrow$ bottom	+0.006	0.000	+0.023
help $\rightarrow$ top	-0.017	-0.180	-0.024
-hurt $\rightarrow$ top	-0.000	-0.120	-0.023
random $\rightarrow$ top (3-direction mean)	-0.041	-0.193	-0.254
hurt $\rightarrow$ top	-0.112	-0.460	-0.663
-help $\rightarrow$ top	-0.105	-0.440	-0.655

Table 8: Fresh-baseline causal steering at 20% strength on 50 usable held-out ballots. Values are mean changes relative to the paired unsteered baseline; allocation cells use finite normalized or natively valid pairs.

A compact regression check summarizes the full grid (Table 9). Define signed semantic strength as $+s$ for help and negated-hurt interventions, $-s$ for hurt and negated-help interventions, and zero for random directions. Define random strength as s for matched-norm random directions and zero otherwise. Regressing target allocation change on these variables, baseline allocation, and a top-policy indicator gives a large positive coefficient for signed semantic strength (+0.182, cluster SE 0.049), while random strength is small and not significant (-0.028, SE 0.039). The effect is stronger for top targets (+0.233, SE 0.080) than bottom targets (+0.133, SE 0.062), although both estimates are positive. Thus the semantic directions carry information beyond perturbation strength, but the effect is about half the exploratory estimate.

Regression	Term	Coef.	Cluster SE	n
Target Δ allocation	signed semantic strength	+0.182***	0.049	1873
Target Δ allocation	random strength	-0.028	0.039	1873
Target Δ allocation	baseline allocation	-0.414***	0.111	1873
Target Δ allocation	top-policy target	+0.177**	0.063	1873
Valence crossing	strength	+0.021	0.087	1200
Top-target spillover	target Δ allocation	-0.312***	0.065	302

Table 9: Regression checks for the fresh-baseline steering run, with standard errors clustered by ballot (50 clusters). The valence-crossing model is a linear probability model over semantic interventions. The spillover model uses 20%-strength top-policy interventions and non-target fresh-rank-2/3 candidates. ** and *** denote $p < 0.01$ and $p < 0.001$.

Valence crossing is weaker than in the exploratory run. At 20%, intended crossing rates are 8% for help-to-bottom, 10% for negated-hurt-to-bottom, 10% for hurt-to-top, and 8% for negated-help-to-top, versus 4.0% and 4.7% for the random bottom and top means. However, crossing does not increase with strength in the regression (+0.021, SE 0.087). We therefore do not treat sign crossing as replicated dose-response evidence.

Finally, the fixed budget mechanically requires redistribution, so we ask whether that redistribution is structured. At 20% strength, hurt-to-top lowers the target by -0.112 while non-target middle candidates gain +0.098 in aggregate; negated-help-to-top lowers the target by -0.105 while middle candidates gain +0.065. A per-ballot spillover regression confirms the pattern: for top-policy interventions, a one-point decrease in the target’s allocation predicts a +0.312 increase in the summed allocation of non-target middle candidates (cluster SE 0.065). This characterizes where the expressed influence moves; it is not evidence that candidate utilities are independent before the budget is imposed.

7 Discussion

Absolute Allocation is useful here because it turns LLM evaluation into a behavioral assay. It supplies labels that best-pick and weak ordinal ballots do not: help, hurt, neutrality, signed mass, pairwise allocation preference, and intensity. The central picture is not that LLM evaluators contain a stable, linearly readable cardinal utility. The evidence points to a narrower and more interesting mechanism:

candidate activations $\rightarrow$ comparative preference $\rightarrow$ weak ordinal judgment
 $\rightarrow$ context-conditioned allocation output.

Candidate activations encode enough information to predict winners, help, hurt, and pairwise preference. But the extra scalar intensity beyond the voter’s own ordinal ballot is not linearly recoverable. The cents look cardinal-looking. We find linear evidence for comparative ordering and polarity, but no corresponding linear evidence for cardinal intensity beyond the generated ordinal ballot.

The fresh-baseline steering result suggests that the bipolar signal is also writable, but less cleanly than the exploratory run implied. Signed semantic strength predicts allocation movement beyond matched-norm random strength, especially when promoting bottom targets or demoting top targets. This does not contradict the residual-intensity null: a direction can causally alter a comparative signal even when exact scalar magnitudes are not independently linearly readable. The absent crossing dose response and dependence on normalized ballots argue against interpreting the intervention as direct control of a stable cardinal utility.

The cross-model results bound this mechanistic claim. Llama produces a rich bipolar label distribution after normalization, but only 14% of attempted ballots are natively valid and its native subset contains almost no hurt labels. Gemma is highly compliant yet rarely expresses opposition at all. The readable axis is robustly established for Qwen; steerability remains Qwen-specific and conditional on deterministic recovery of most intervention ballots.

Context matters throughout, but its form is model-specific. Exact-paired reference display changes Llama best-pick winners on 27–29% of prompts while changing Borda and Absolute Allocation winners on about 10–12%. It shifts Llama from neutrality toward support, whereas the less tightly paired Qwen comparison shifts toward opposition. Presentation order produces a strong Qwen primacy gradient but little Llama primacy. Random activation perturbations are not harmless. These are limitations for LLM juries, but they are exactly the artifacts Absolute Allocation makes measurable.

8 Limitations and Future Work

This is an early prototype. The prompt set is small, the candidate generator is fixed, and the main external validator is still an LLM rather than a human gold standard. The behavioral validation and voter-level representation runs use related but distinct conditions; future work should unify them in a single larger experiment.

The Qwen reference-display comparison reuses exact self-answers and presentation orders and includes a same-seed hidden placebo. However, 17 invalid anchor contexts lacked archived display mappings and were omitted from both paired runs, so estimates condition on anchor ballot validity. The Llama comparison also reuses exact inputs but has much lower native-validity overlap; its best-pick result is stronger than its exact allocation-movement result.

The ballot prompt itself is another limitation. Its formatting example uses the real candidate labels and demonstrates A as the winner with a strict $A > B > C > D$ ranking and a one-hot allocation. The strong A preference in Gemma and Llama may partly reflect this anchor. Shuffling presentation order does not break the association between the example and candidate identity. Future cross-model work should use dummy example IDs, permute labels independently of content, or use constrained decoding without a substantive example.

The ordinal ballot is a limitation. Because ranked groups were allowed, the Borda-style ballot is a weak ranking with ties rather than canonical strict Borda. A clean comparison against strict Borda requires rerunning the ordinal ballot with no ties allowed. The ballot schema also elicits best-pick, then the ranking, then the allocation, so allocations are generated conditioned on the ranking just produced. The primary experiment randomizes field order per ballot and records it;

field order alone is at chance and adding it to the activation probes does not materially change performance. Separate field-specific analyses remain useful because randomization estimates an average across elicitation orders rather than eliminating order effects.

Two further confounds. First, answer length: the four candidate styles differ systematically in length and verbosity is a known judge bias, but a length-matched control (script `level25_length_controls.py`) shows length alone is only weakly predictive (help/hurt/allocation-top AUC 0.57, best-pick 0.52) and that decodability survives length matching (pooled within-length AUC 0.62–0.77, all strata above chance), with the residual-beyond-rank null unchanged when length is added to the baseline; the steering analysis remains uncontrolled for length. Second, the behavioral reliability of the allocation cents is unmeasured: despite the no-repair probe sensitivity, the residual-intensity null presupposes a reliability we have not established, and a test–retest design that resamples ballots at the generation temperature would show whether intensity is even reproducible before asking whether it is decodable.

The steering experiment remains limited to one model, one preselected layer, 50 usable intervention ballots, three random directions, and a maximum intervention strength of 20% of the median activation norm. The rerun improves the design by selecting targets from the fresh paired baseline, replaying the exact randomized-field protocol, and measuring baseline-generation noise. Its dominant limitation is ballot validity: only 10.1% of steered generations are natively exact-budget compliant, 79.0% are recovered by deterministic normalization, and 10.8% are unusable. Results therefore depend on a transparent but consequential measurement rule and a recoverable-ballot subset. Replication should use constrained decoding or a simpler schema to obtain natively valid ballots, then repeat across layers, models, and larger held-out sets.

The activation analyses are observational except for the steering intervention, and linear probes do not establish a full circuit. Larger prompt sets, non-Qwen mechanistic replications with sufficiently balanced natively valid labels, nonlinear neutrality probes, human spot checks, and causal tests for position or reference debiasing are natural next steps.

9 Conclusion

Absolute Allocation exposes a useful dissociation in Qwen evaluation. The expressed ballot contains polarity and cardinal-looking magnitude beyond weak ranking, while candidate activations provide linear evidence principally for comparative ordering and bipolar polarity. Semantic probe directions causally influence those expressed judgments beyond matched-norm random strength, but the intervention does not establish stable cardinal control and depends on normalization of most usable outputs. Reference display and position show that the readout is context-sensitive; Llama and Gemma show that the current prompt does not elicit a measurement-invariant object across model families. The contribution is therefore not a new voting rule or a proof that intensity is constructed at generation time. It is a signed behavioral assay that separates what an evaluator expresses from what is linearly readable and behaviorally steerable in its candidate representations.

References

- [1] Rui Ai, Yuqi Pan, Milind Tambe, and Haifeng Xu. Beyond majority voting: Llm aggregation by leveraging higher-order information. *arXiv preprint arXiv:2510.01499*, 2025.
- [2] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Cather-

- ine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Tom Telleen-Lawton, Tom Brown, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Jared Brown, and Jared Kaplan. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022.
- [3] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint arXiv:2305.14325*, 2023.
 - [4] John Connor Sullivan. The absolute allocation voting model: Ma thesis. Zenodo 10.5281/zenodo.21286283, 2025.
 - [5] Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.
 - [6] Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. *arXiv preprint arXiv:2305.17926*, 2023.
 - [7] Weiqin Wang, Yile Wang, and Hui Huang. Ranked voting based self-consistency of large language models. *Findings of the Association for Computational Linguistics: ACL 2025*, 2025.
 - [8] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.
 - [9] Shaolei Zhang, Tian Zhang, Yang Yang, and Minlie Huang. Aligning large language models with human preferences through representation engineering. *arXiv preprint arXiv:2312.15997*, 2023.
 - [10] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*, 2023.
 - [11] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023.